Lab 2D: Package and Ship — Plugins & Marketplaces

Duration: 35 min **After:** Plugins & Marketplaces lecture (Day 2, Block 4) **Prerequisites:** Labs 2A–2C complete (skills and hooks in `.claude/`)

Objective

Inspect a marketplace entry before installation, scaffold a skills-directory plugin with `claude plugin init`, validate it strictly, publish the proven package through a local marketplace, and install it back at an intentional scope.

Deliverable (toolkit chain 8 of 12)

A validated `dashboard-toolkit` plugin containing two skills and two hooks, plus a local marketplace that can distribute it.

> ***START MARKER:*** *`/clear` done; `/skills` shows `code-review` and `review`; both hooks passed Lab 2B.*

Exercise 1: Inspect Before Installing (6 min)

Open the plugin manager:

```
/plugin
```

Choose a plugin from `claude-plugins-official`, open its detail view, and inspect the publisher plus every contributed component. The official marketplace is a catalog; entries can have different publishers and maintenance models.

Use this review checklist:

1. Publisher, source repository, release activity
2. Skills, commands, agents, hooks, scripts, MCP/LSP servers, dependencies
3. Files, commands, credentials, and user privileges it can access
4. Network destinations, model/provider changes, and stored state
5. Version/update policy plus disable, uninstall, and rollback path

Record an **APPROVE**, **REJECT**, or **APPROVE WITH CONDITIONS** decision with one sentence of evidence.

> *Plugins are executable dependencies. Marketplace placement is not a security boundary.*

Exercise 2: Scaffold the Plugin (6 min)

Exit Claude and run:

```
claude plugin init dashboard-toolkit \
  --description "Review skills and deterministic enforcement hooks" \
  --with skills hooks
```

Current behavior: the command creates `~/.claude/skills/dashboard-toolkit/` and loads it as `dashboard-toolkit@skills-dir` on the next session. It does not scaffold in the current directory.

Inspect and validate:

```
claude plugin validate ~/.claude/skills/dashboard-toolkit --strict
claude plugin list
```

Expected output marker: strict validation succeeds and the plugin appears from the `skills-dir` source.

Exercise 3: Package the Toolkit (8 min)

Start Claude in `business-dashboard` and run:

```
Update ~/.claude/skills/dashboard-toolkit so it contains:
1. My .claude/skills/code-review/ skill as skills/code-review/
2. My .claude/skills/review/ human-invoked skill as skills/review/
3. My post-edit-format.sh and pre-write-check.sh scripts under hooks/
4. hooks/hooks.json declaring the PostToolUse Write|Edit formatter and PreToolUse Write
blocker using ${CLAUDE_PLUGIN_ROOT}-relative paths
5. A manifest description that names the skills and hooks
Remove placeholder example components that the scaffold created.
Do not add an MCP server, external network dependency, credential, or provider override.
```

Validate again:

```
claude plugin validate ~/.claude/skills/dashboard-toolkit --strict
claude plugin details dashboard-toolkit@skills-dir
```

Expected output marker: validation passes and the component inventory matches the intended package. Review the inventory before continuing.

Exercise 4: Publish Through a Local Marketplace (10 min)

Create this structure next to `business-dashboard`:

```
team-marketplace/
├── .claude-plugin/
│   └── marketplace.json
└── plugins/
    └── dashboard-toolkit/
        ├── .claude-plugin/plugin.json
        ├── skills/
        └── hooks/
```

Copy the validated plugin into `team-marketplace/plugins/dashboard-toolkit/`. Create `.claude-plugin/marketplace.json`:

```json
{
  "name": "team-marketplace",
  "owner": { "name": "Training Team" },
  "plugins": [
    {
      "name": "dashboard-toolkit",
      "source": "./plugins/dashboard-toolkit",
      "description": "Review skills and deterministic enforcement hooks"
    }
  ]
}
```

From `business-dashboard`, run:

```
/plugin marketplace add ../team-marketplace
/plugin install dashboard-toolkit@team-marketplace
```

Choose **local scope** for the classroom smoke test unless the instructor explicitly asks you to share the project configuration.

Then verify:

```
claude plugin list
claude plugin details dashboard-toolkit@team-marketplace
```

If the install summary requests activation, run:

```
/reload-plugins
```

Test `/dashboard-toolkit:review server.js` and verify both hooks on safe test files.

Exercise 5: Prove Reversibility (5 min)

```
/plugin disable dashboard-toolkit@team-marketplace
/plugin enable dashboard-toolkit@team-marketplace
/plugin uninstall dashboard-toolkit@team-marketplace
```

Reinstall at local scope after proving the lifecycle. Record which settings file changed for the selected scope.

FINISH MARKER — Success Checklist

- [ ] Inspected exact publisher, components, authority, data flow, and lifecycle before installation
- [ ] Recorded an evidence-based trust decision
- [ ] `claude plugin init dashboard-toolkit --with skills hooks` created a skills-directory plugin
- [ ] `claude plugin validate ... --strict` passed
- [ ] Marketplace uses `./plugins/dashboard-toolkit` inside its own repository
- [ ] Plugin installed at intentional scope and activated successfully
- [ ] Namespaced review skill and hooks work
- [ ] Disable, enable, uninstall, and reinstall paths were proven

If Stuck

Problem	Recovery
Scaffold not in current folder	Expected: it is under `~/.claude/skills/dashboard-toolkit/`
Strict validation fails	Fix every schema error or warning before publication
Marketplace source not found	Keep plugin under `team-marketplace/plugins/` and use a marketplace-relative source
Plugin installed but inactive	Follow the install summary; run `/reload-plugins` when requested
Old component remains	Remove the installed copy, update the marketplace, and reinstall at the intended scope
Hook path breaks after install	Use `${CLAUDE_PLUGIN_ROOT}` in plugin hook paths

Debrief Questions

1. Why is a marketplace a distribution mechanism rather than a security boundary?

2. Which component in your plugin carries the most authority?
3. What evidence would your organization require before moving this plugin into an approved internal marketplace?